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Model Context Protocol (MCP) Gateway — a middleware meant to productionize MCP for an enterprise

6 min read · Jun 13, 2025



Manoj Jahgirdar

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Learn how ContextForge MCP Gateway works — a secure, unified middleware for scaling agentic AI integrations in the enterprise.

Introduction

Model Context Protocol (MCP) launched by Anthropic has become a goto standard to integrate tools in Agentic AI applications and has significantly simplified agent-to-tools communication. However, when we talk about system design for and enterprise Agentic AI, we need to take care of consistency, security, observability, and more. Just like how every enterprise systems require an API Gateway that supports rate limiting, SSL termination, authentication, IP whitelisting, etc we need a MCP Gateway for an Agentic AI application to communicate with MCP server.

What is MCP Gateway?

ContextForge MCP Gateway is an open-source reverse-proxy that unifies MCP and REST tool servers under a single secure HTTPS endpoint with discovery, auth and observability baked in.

What does it do?

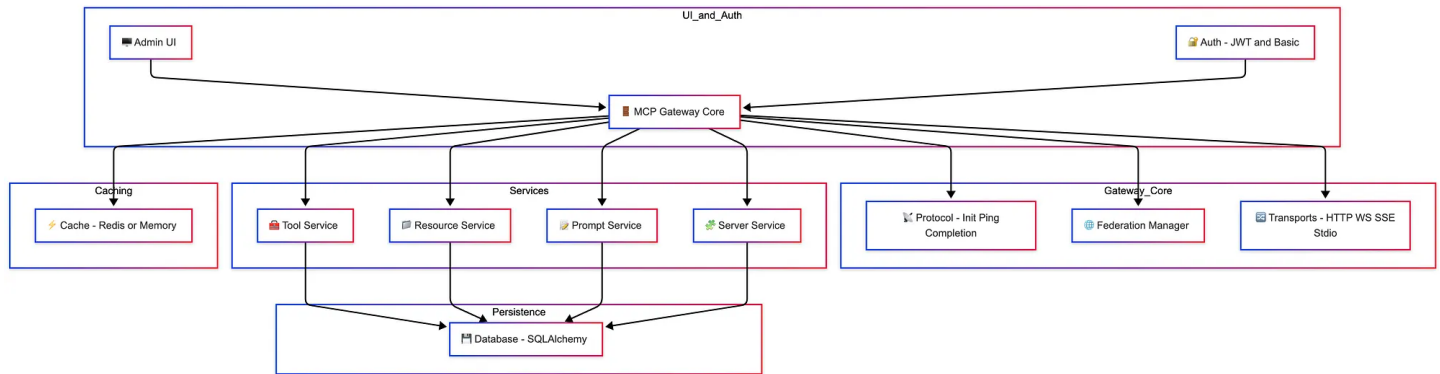
It acts as a unified interface for tools, resources prompts, virtual servers and federated gateways. It has a multi-transport support meaning it can be accessible through transport protocols such as HTTP/JSON-RPC, WebSocket, Server-Sent Events(SSE) & stdio.

Core components of MCP Gateway

MCP Gateway consists of the following components,

1. **Virtual Servers:** A virtual or a federated server is a wrapper that combines selected tools, prompts and resources under one context-specific endpoint. And this server can be exposed through **Stdio** or **SSE** as an MCP server.
2. **Server Catalog:** A collection of virtual servers discoverable through REST endpoints.
3. **Tools:** Tools are remote functions that an LLM can invoke, either via MCP or REST. Native or REST-adapted tools are supported. You can wrap any REST API, CLI command to be a tool.
4. **Prompts:** Prompts are reusable message templates with arguments. They define system prompts, user instructions, or chainable inputs. *Jinja2 template* with schema validation, versioning and rollback are supported.
5. **Resources:** Resources expose read-only data like files, database rows, logs, or screenshots. LLMs can read this content through an URI. MIME-aware, URI-addressable, LRU+TTL caching (redis in-memory or DB) & SSE based subscription to dynamic resources are supported.
6. **Roots:** Roots define base folders for file-based resources. They control what files MCP clients can access from your local system.
7. **Metrics:** Track tool calls, resource reads, prompt renders, and overall usage in one place. Structured JSON logs and `/health` endpoint with live latency stats are supported.
8. **Gateway:** Gateways are other MCP-compatible servers. When registered, their tools/resources/prompts become usable locally.

9. **Authentication & Security:** Supports both Basic and JWT authentication. It also supports TLS verification options.
10. **Admin UI:** MCP Gateway includes a built-in Admin UI for managing all entities in real time via a web browser.



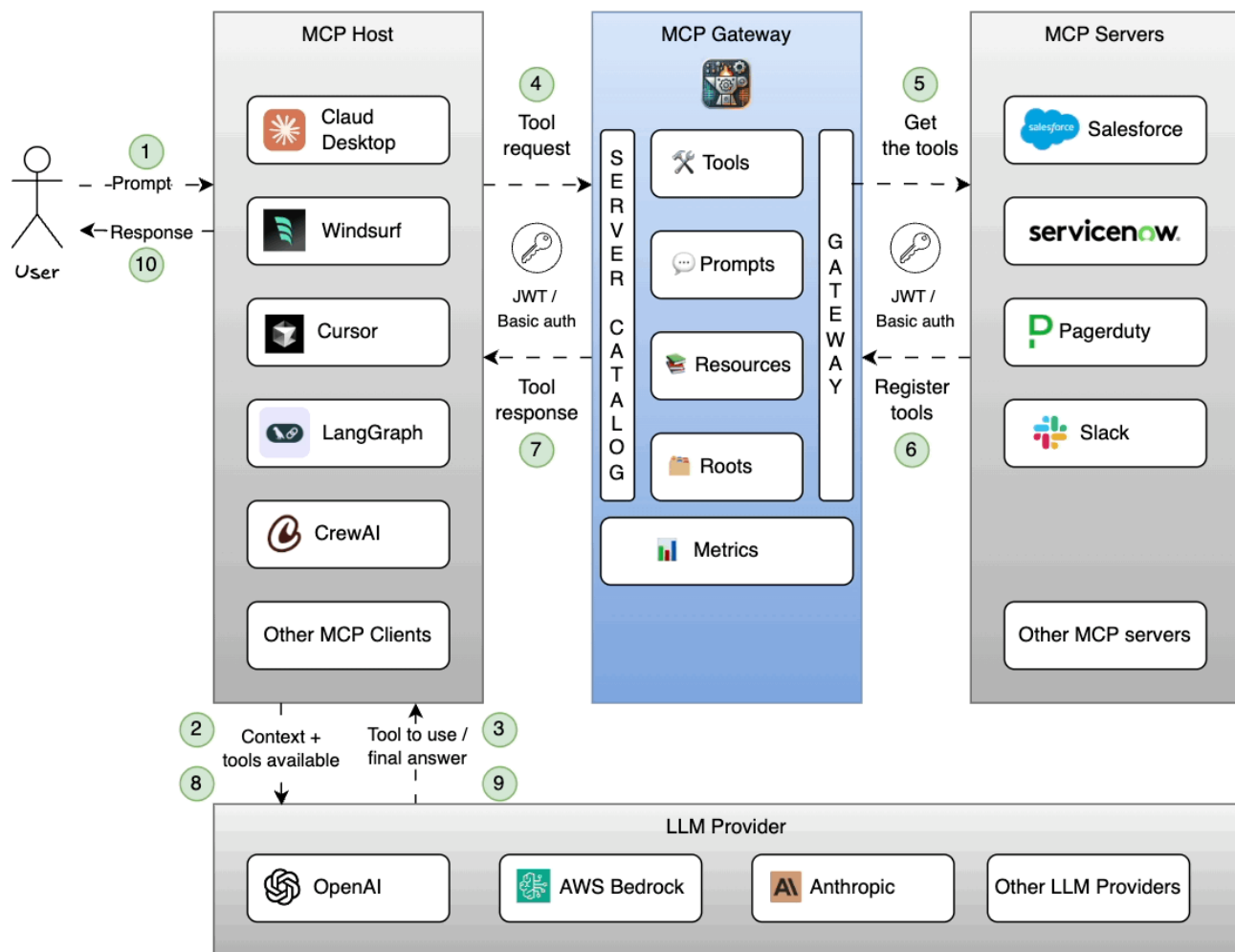
MCP Gateway architecture

How does it work?

I'll explain the importance and working of MCP Gateway in an enterprise Agentic AI solution with two real world use cases:

1. **Automated IT Incident Navigation:** On-call engineers can have a single chat interface to diagnose, triage and remediate high-severity incidents.
2. **Automated Opportunity Intelligence Assistant:** Sales leaders can have a single chat interface that instantly synthesizes every piece of account data and recommends next steps.

Reference Architecture



MCP Gateway implementation

Use case 1: Automated IT Incident Navigation

On-call engineers today struggle with fragmented dashboards — PagerDuty for alerts, ServiceNow for tickets, CMDB for asset context and Slack for collaboration — forcing them to manually stitch together logs, recent changes and static runbooks under pressure. This disjointed process leads to slow triage, missed remediation steps, and extended outages that breach SLAs and frustrate customers.

With automated IT Incident Navigation system, On-call engineers can have a single chat interface to diagnose, triage and remediate high-severity incidents faster.

Flow:

1. On-call engineer enters a prompt *“Help me resolve P1 incident #1234”* in the MCP

Client lets say Cloud Desktop.

2. Assuming that the MCP client is already registered with an MCP Gateway, the client sends this context and the tools available to the LLM.
3. LLM reasons through and gives which tool(s) needs to be invoked and the arguments to invoke the tool(s).
4. MCP client authenticates with the Virtual server in the MCP Gateway through JWT / Basic authentication mechanism for a secure session. A virtual server has be created in MCP Gateway with selected tools, prompts and resources for this use case under one context-specific endpoint.
5. The external MCP servers such as PagerDuty, ServiceNow are accessed through the Gateway.
6. The tools from the external MCP servers are registered in Gateway. Tools such as `getIncident` from PagerDuty, `getCMDBContext` and `listTickets` from ServiceNow.
7. The tools are executed and the response is sent back to MCP client.
8. MCP client sends the he tool responses to the LLM as a context.
9. The LLM reasons through and arrives at a final answer that includes: A natural-language incident summary and a ranked list of next-steps (e.g. “rollback deploy,” “scale pods,” “run database health check”)
10. Client displays the full conversation, logs metrics on resolution time, and closes the loop when the incident’s resolved.

Optional enhancements:

- Optionally kick off a Rundeck playbook via PagerDuty or ServiceNow runbook API.
- Posts the summary + actionable buttons (Acknowledge, Escalate, Run Playbook) into the on-call Slack channel via the Slack tool.

Benefits:

- One unified prompt instead of hopping between UIs.
- AI-backed triage + automated remediation.
- Full audit trail across PagerDuty, ServiceNow, Slack.

Use case 2: Automated Opportunity Intelligence

Sales teams lack a unified view of deal health because CRM data (Salesforce), IT provisioning status (ServiceNow) and team discussions (Slack) all live in separate silos. Reps spend up to an hour before each customer call gathering context — deal size, contract clauses, outstanding support tickets and internal chatter — which causes them to miss critical risk signals and upsell opportunities until it's too late.

With Automated Opportunity Intelligence Assistant, Sales leaders can have a single chat interface that instantly synthesizes every piece of account data and recommends next steps.

Flow:

1. Similar to the above flow, the user can ask something like *“What’s the health of Acme Corp opportunity XYZ-789?”*.
- 2, 3. remain the same.
4. A virtual with different set of tools, prompts and resources are targetted here.
5. remains same.
6. Tools such as `einsteinGPT` from Salesforce for opportunity stage, deal size, risk factors and `assetProvisionTicket` from ServiceNow pulls any asset-provision tickets.
- 7, 8. remains the same.
9. The LLM reasons through and arrives at a final answer that includes: A one-paragraph executive summary (“Deal is 60% likely; blocking issue on network provisioning still open in SNOW; C-level sponsor wants PO by EOD”) and A 3-item playbook (“1. Get provisioning ticket to ‘In Progress’; 2. Send weekly exec update; 3.

Upsell training add-on”)

10. remains same.

Optional enhancements:

- Creates or updates a ServiceNow project task for provisioning.
- Posts the summary and playbook as a threaded message in “#acme-xyz” Slack.

Benefits:

- Instant 360° view across CRM, ITSM, and collaboration data.
- AI-driven next-best-actions.
- Seamless follow-up without context loss.

Conclusion

I explained the importance of having an MCP Gateway in building your enterprise Agentic AI solution through two real-world examples.

Try the ContextForge MCP Gateway here:

- GitHub repo: <https://github.com/IBM/mcp-context-forge>
- Documentation: <https://ibm.github.io/mcp-context-forge>

Call-to-action for GitHub ★ — leave a star and consider contributing to our good first issue: [Create an issue](#).

Additional resources

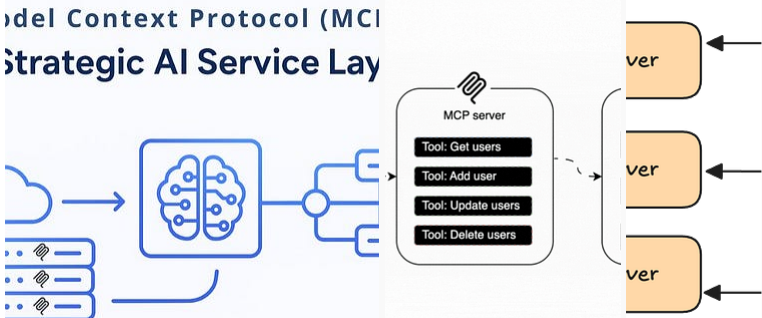
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Written by Manoj Jahgirdar

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I am a Software Engineer with over 6 years of experience in IT Industry. I write about enterprise grade systems and solutions. Scalability, Resilience, and more

Responses (1)



Dinesh Kumar

What are your thoughts?



Mmeghna
Oct 19



Great read.

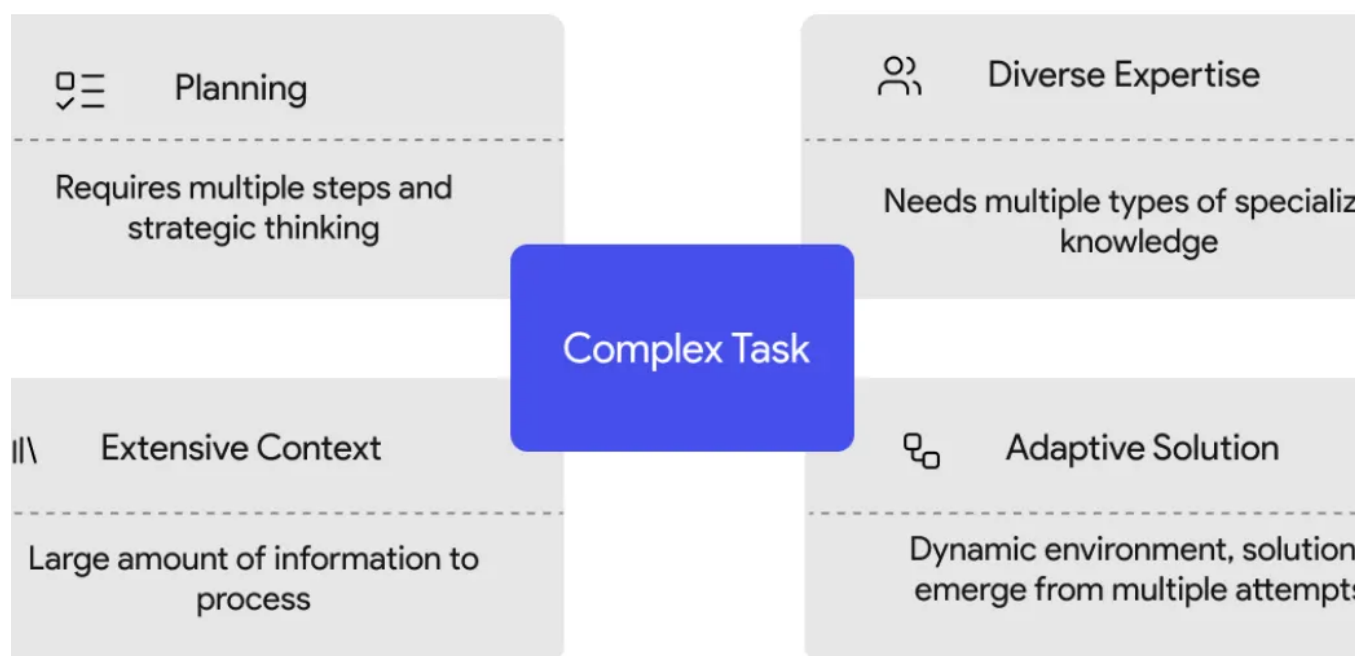
Would have loved to understand more on the messages exchanged between the MCP Gw and the server when registering and Client



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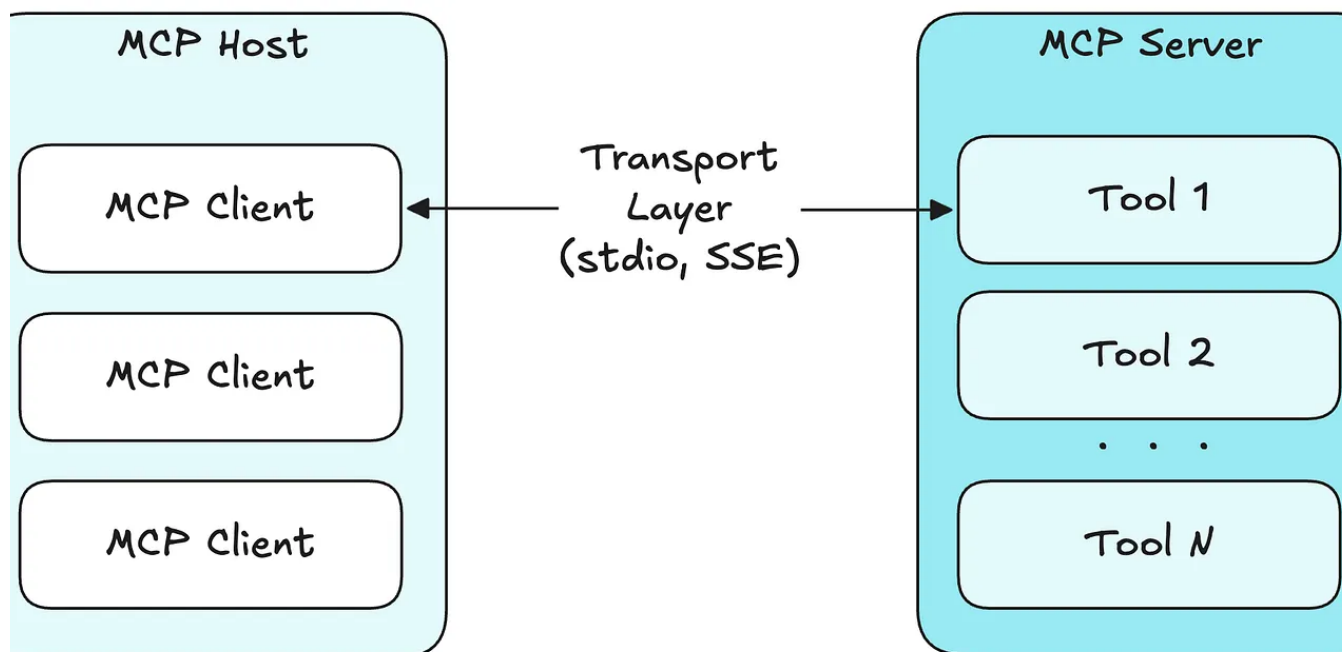


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Manoj Jahgirdar

Developing a Model Context Protocol (MCP) server and client for your agent-tool interoperability

In this blog, you will learn to implement MCP server and MCP client.



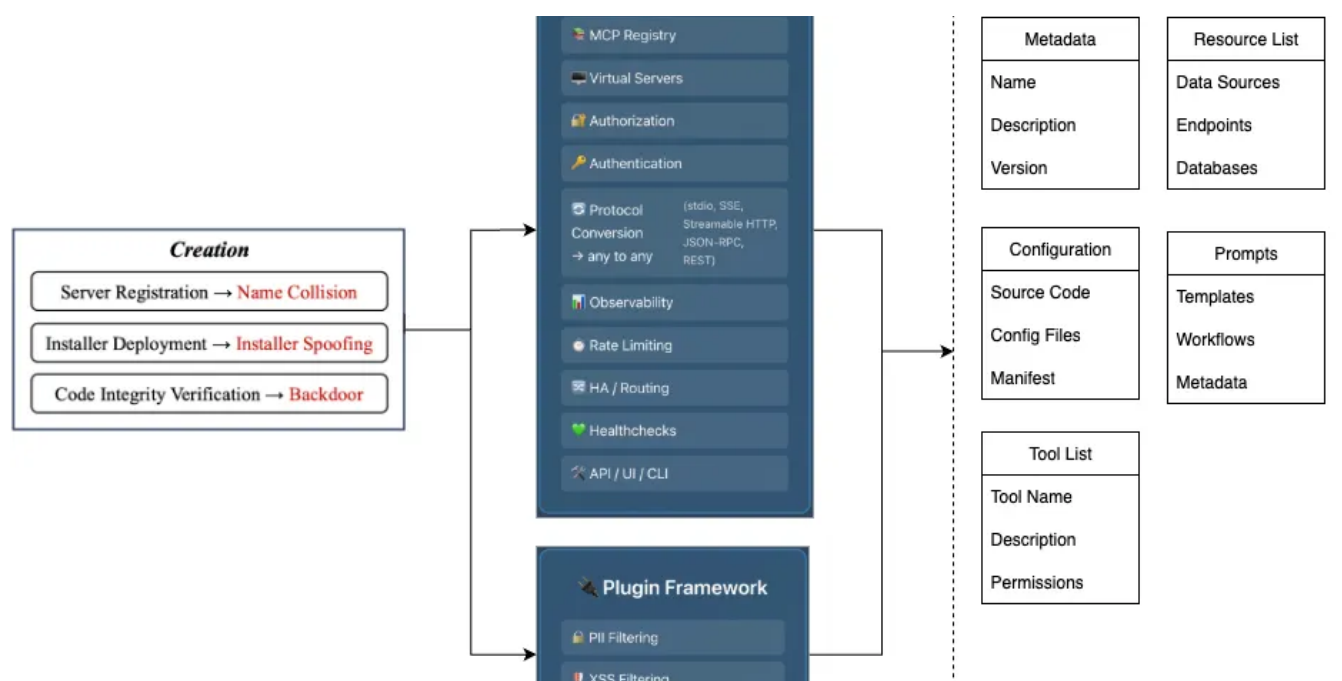
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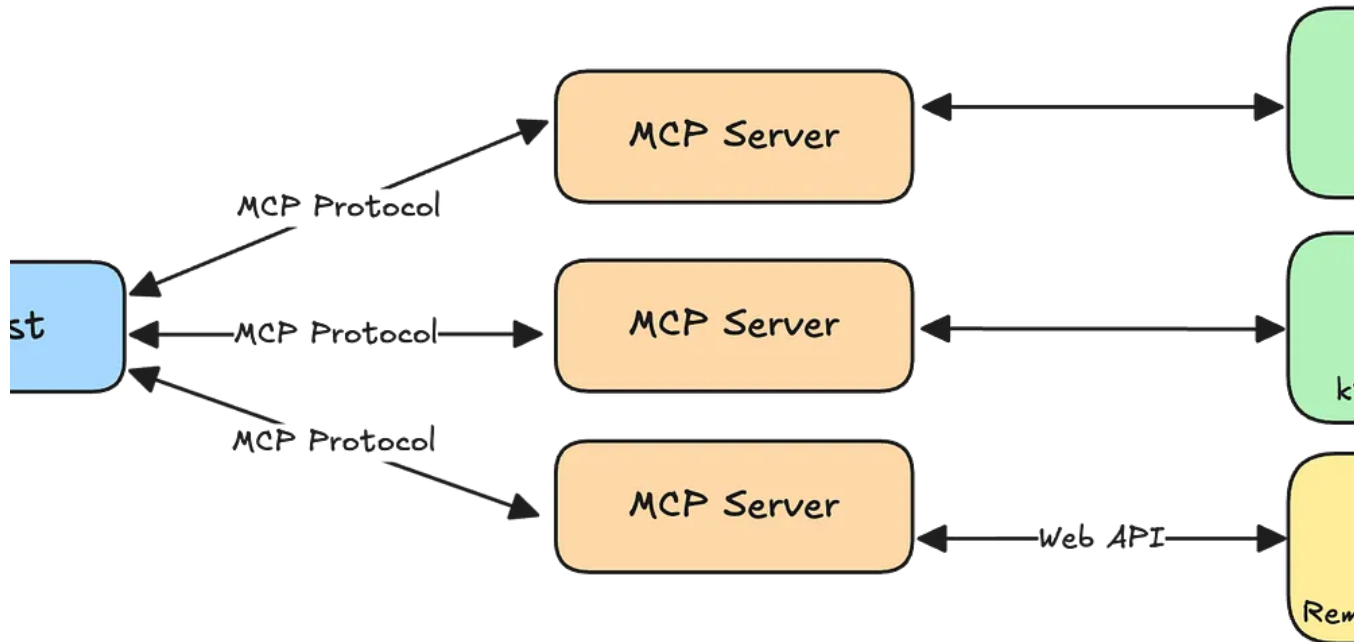


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A deep dive into securing AI agent ecosystems with ContextForge MCP Gateway at the heart of enterprise-grade protection.

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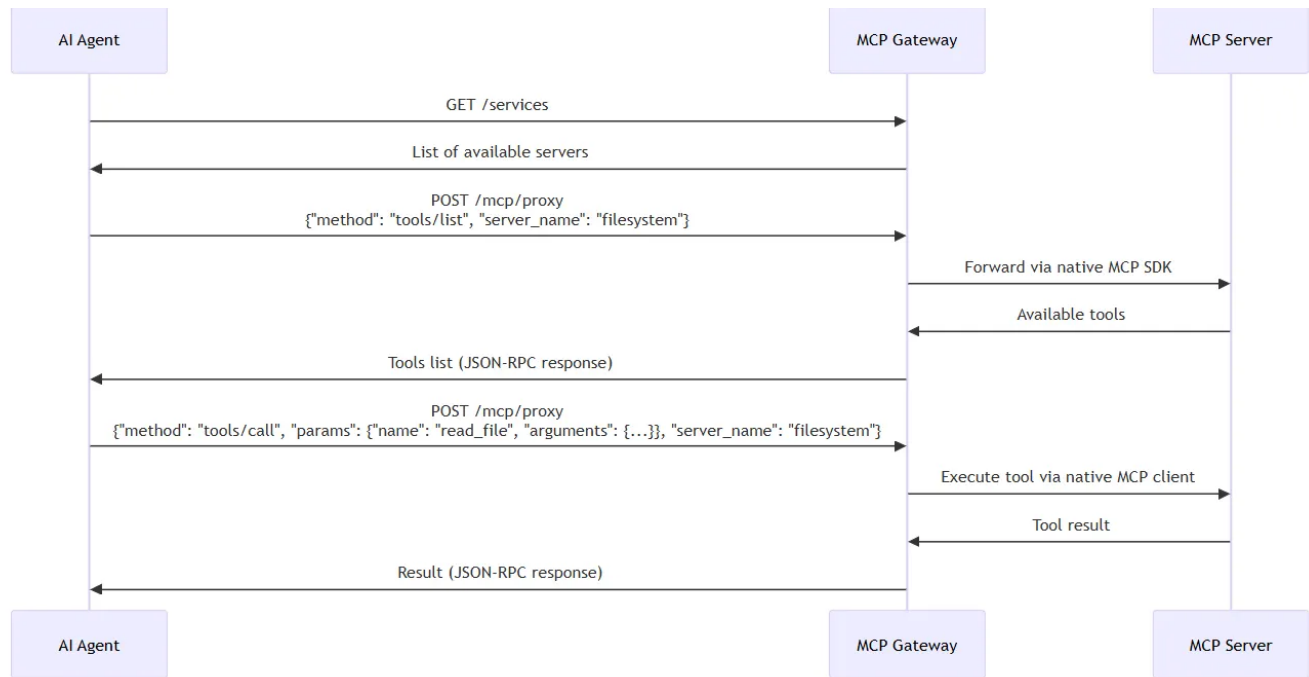
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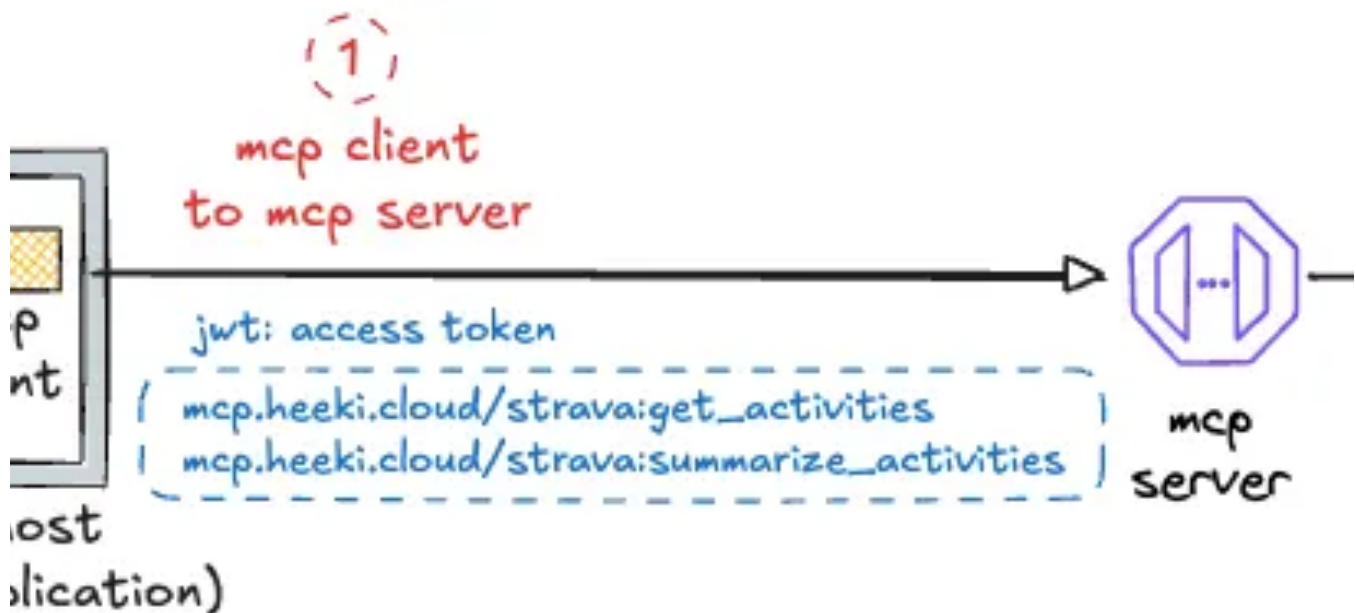


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Tools

List Tools

Clear

Select a tool

Select a tool from the list to view its details

8000/sse

📄 Servers File

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🔌 Disconnect

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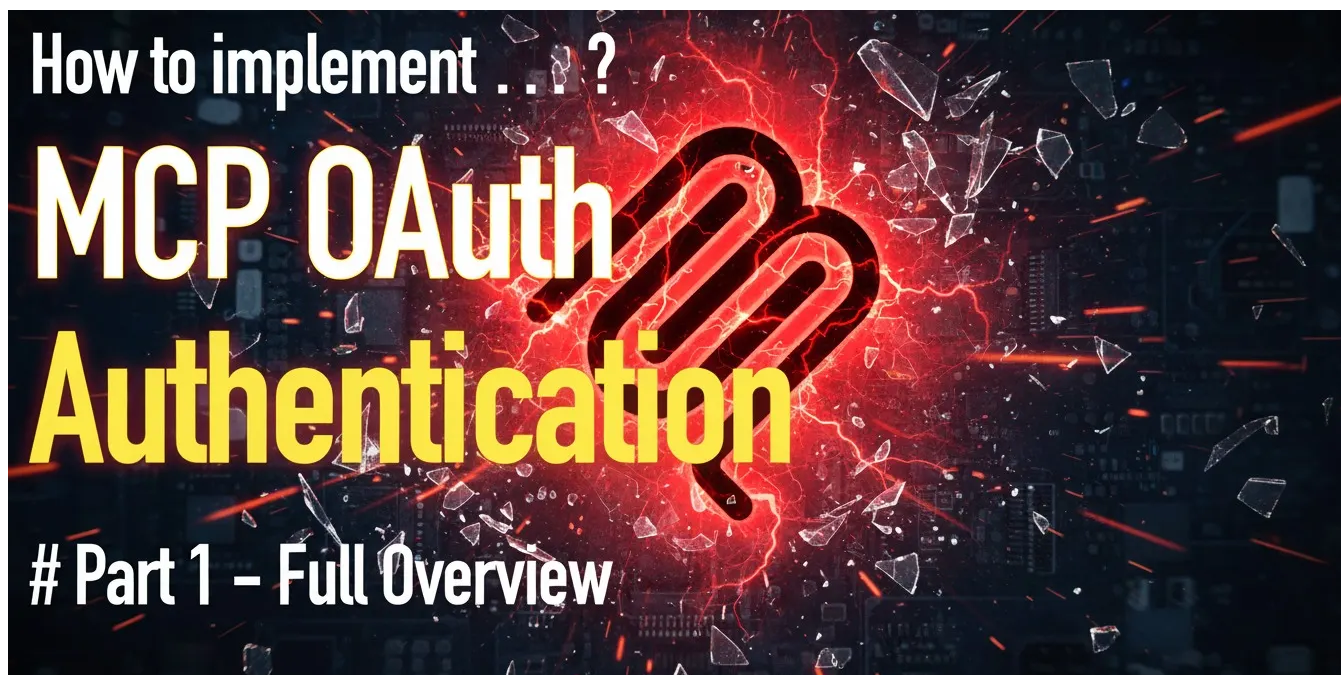
Xin Cheng

MCP Server OAuth Protection: Part 1

Secured MCP Server

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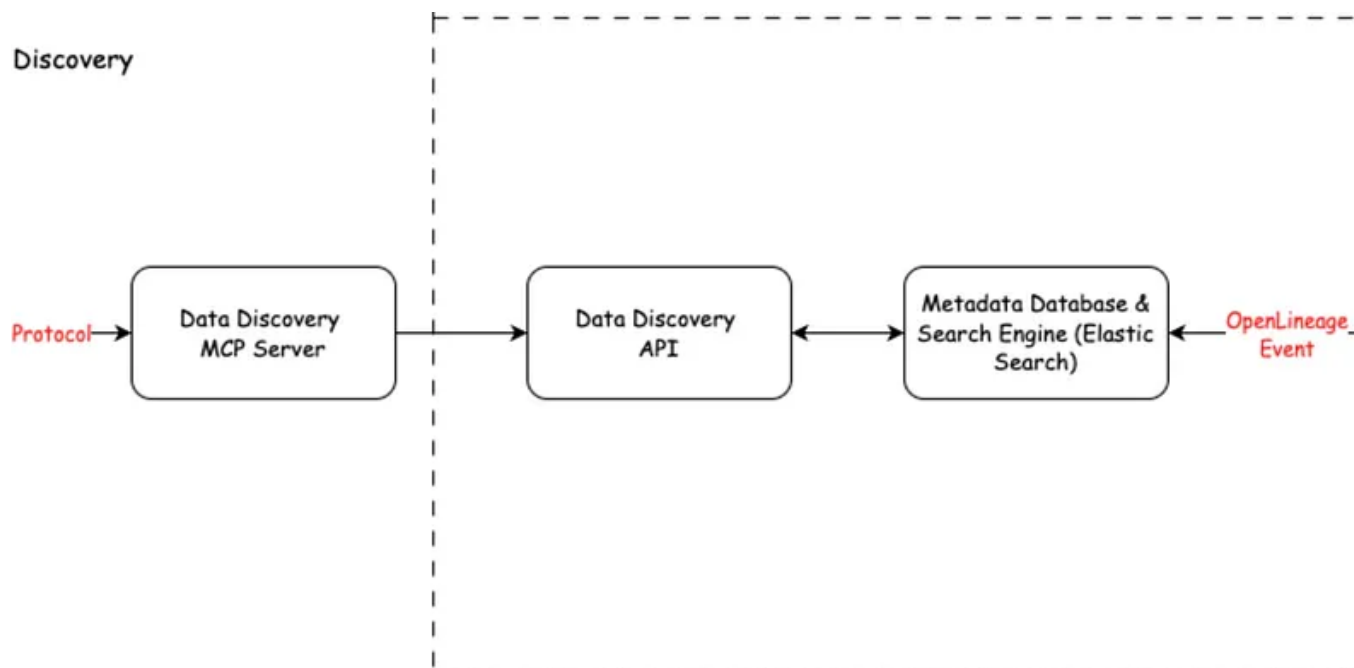


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Liangjun Jiang

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